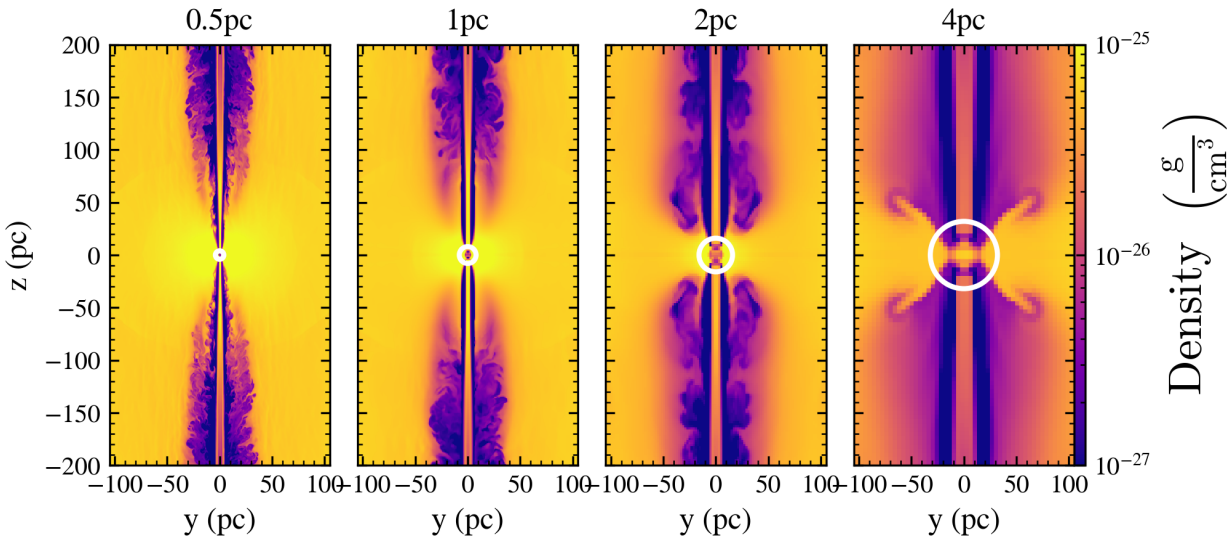

Figure 11. Gas density slice at $t = 0.2$ Myr for JA runs.

In the end, enthalpy is lower for higher resolution runs due to a more confined jet.

Based on the above observations, we conclude that increasing resolution can lead to a better convergence of the jet feedback. The cell size of 1 pc ($0.5r_{\rm B}$) is the least acceptable resolution to achieve high fidelity in these jet feedback simulations. This means the control surface radius also needs to be less than half of $r_{\rm B}$ for more physical runs.

3.2.2. *Size of Jet Injection Region*

The height $h_{\rm jet}$ and radius $r_{\rm jet}$ of the jet injection cylinder are adjustable numerical parameters. Ideally the radius should reflect the typical radius of an AGN jet at the distance represented by the control surface radius. However, jets observed using very long baseline interferometry (VLBI) are still highly collimated at the Bondi radius and may have transverse radii of only a few percent of the Bondi radius at that distance (Blandford et al. 2019). Thus for the jet injection cylinder to be physically correct, we would require the control surface (which is typically about $r_{\rm B}$ or $r_{\rm BHL}$) to be impractically well-resolved. At best we can make the injection cylinder about two zones across. Also, the cylinder radius cannot be larger than the control surface radius $R_{\rm cs}$, or else accretion will only be allowed from the sides.

Unlike the radius of the injection cylinder, its height does not have a physical correlation, since by means of the cylinder we are attempting to impose an inflow boundary condition (i.e. directed from the control surface onto the grid outside it) that determines the mass and energy flux on a two-dimensional surface. The cylinder height must be at least $R_{\rm cs}$ in order to have an effect on gas outside the control surface, and to avoid nonphysically forcing the solution there it should not be much larger than $R_{\rm cs}$.

Given these constraints, we conducted runs for jet feedback with different values of $h_{\rm jet}$ and $r_{\rm jet}$ to explore the sensitivity of the feedback behavior to these parameters. The runs relevant for this section are the JB and JC runs. These runs used a well-resolved control surface with $\Delta x = 0.25$ pc. We kept the control surface radius $R_{\rm cs} = 8\Delta x = 2$ pc and reset radius $R_{\rm reset} = 4\Delta x = 1$ pc. The JB runs vary jet radius, keeping $h_{\rm jet} = 4\Delta x$, while the JC runs keep $r_{\rm jet} = 2\Delta x$ and vary the height of the jet. Slice plots showing the typical jet structure in each run are shown in Figure 16.

Our tests show that $r_{\rm jet}$ is the more important of the two parameters. Regardless of $h_{\rm jet}$, the narrower jets develop a dense spine with a nozzle structure and side shocks. The turbulence within the jet region can vary, but the eventual size of the shocked region varies little among the runs. From the azimuthally and longitudinally averaged profiles of density, pressure, and velocity (Figure 17), we can also see that the pressure and jet velocity do not vary significantly with the height of the jet. The density in the jet spine varies by up to a factor of two with changing $h_{\rm jet}$. The kinetic energy distribution (seen in Figure 13), is also virtually independent of $h_{\rm jet}$. The accretion rate and shocked volume at late times (Figure 14 and 15) also show that varying the width of the jet injection cylinder has a stronger effect than varying its height. At late times, smaller jet heights can result in smaller shocked volume, consistent with what is observed in the azimuthally averaged profile. In

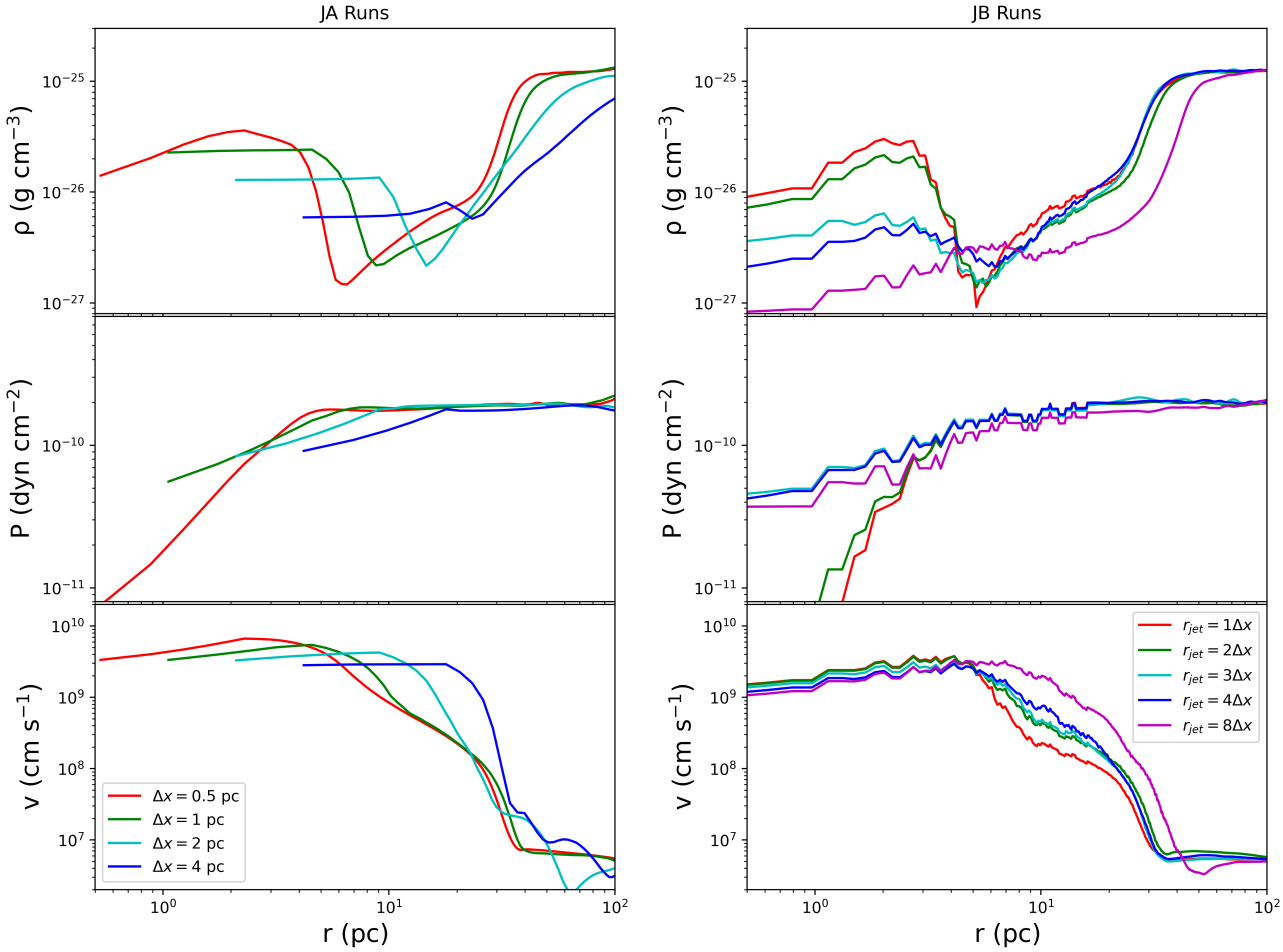

Figure 12. Azimuthally and longitudinally averaged density, pressure, and velocity profiles at $t = 0.2$ Myr for different jet runs. Left: with varying resolution, where control surface and reset radius are scaled with the resolution (JA runs). Right: fixing resolution and varying jet radius (JB runs).

general, we can conclude that the results are relatively insensitive to the height of the jet injection region.

The results are different with varying jet width, however. If $r_{\rm jet} > R_{\rm reset}/2$, the jet injection procedure produces a combination of effects that lead to mass ejections perpendicular to the jet direction. First, a wide jet injection cylinder constricts infalling gas, producing a Venturi effect: material infalling from the side speeds up and develops lower pressure, drawing more material from the side. Second, since the velocity is not modified in the resetting region, the gas remaining after resetting exits the region at high speed, shocking against material falling in from the opposite direction. The result is a pseudo-jet whose direction is determined by random fluctuations in the flow field around the black hole.. This unphysical behavior places an upper limit on acceptable values of $r_{\rm jet}$; to avoid it, the injection cylinder must not overhang the resetting sphere by more than a mesh zone. We also observe that narrower jets form low-density central channels, while wider jets inflate shocks and cocoons with significant size. Their accretion rate and shocked volume show more variation as well, with the smaller injection cylinder allowing less accretion and producing a more slowly expanding transverse shock. Narrower jets also entrain significantly less material.

Based on the above results, a narrow injection cylinder achieves the most realistic results. Observationally, we also know that the jets are narrower even than $0.25R_{\rm cs}$ (in this case 0.5 pc). To achieve a realistic accretion rate according to the simulations, we require $r_{\rm jet}$ to be at least two zones across, and no larger than $R_{\rm reset}/2$.

3.2.3. *SMBH Mass*

We tested the dependence of accretion and jet properties on the central SMBH mass via the set of JD runs. In these runs, we fix the parameters discussed above ($R_{\rm cs}$, $R_{\rm reset}$, $r_{\rm jet}$ and $h_{\rm jet}$) but vary the mass of the accretor, causing the relative proportion of these quantities

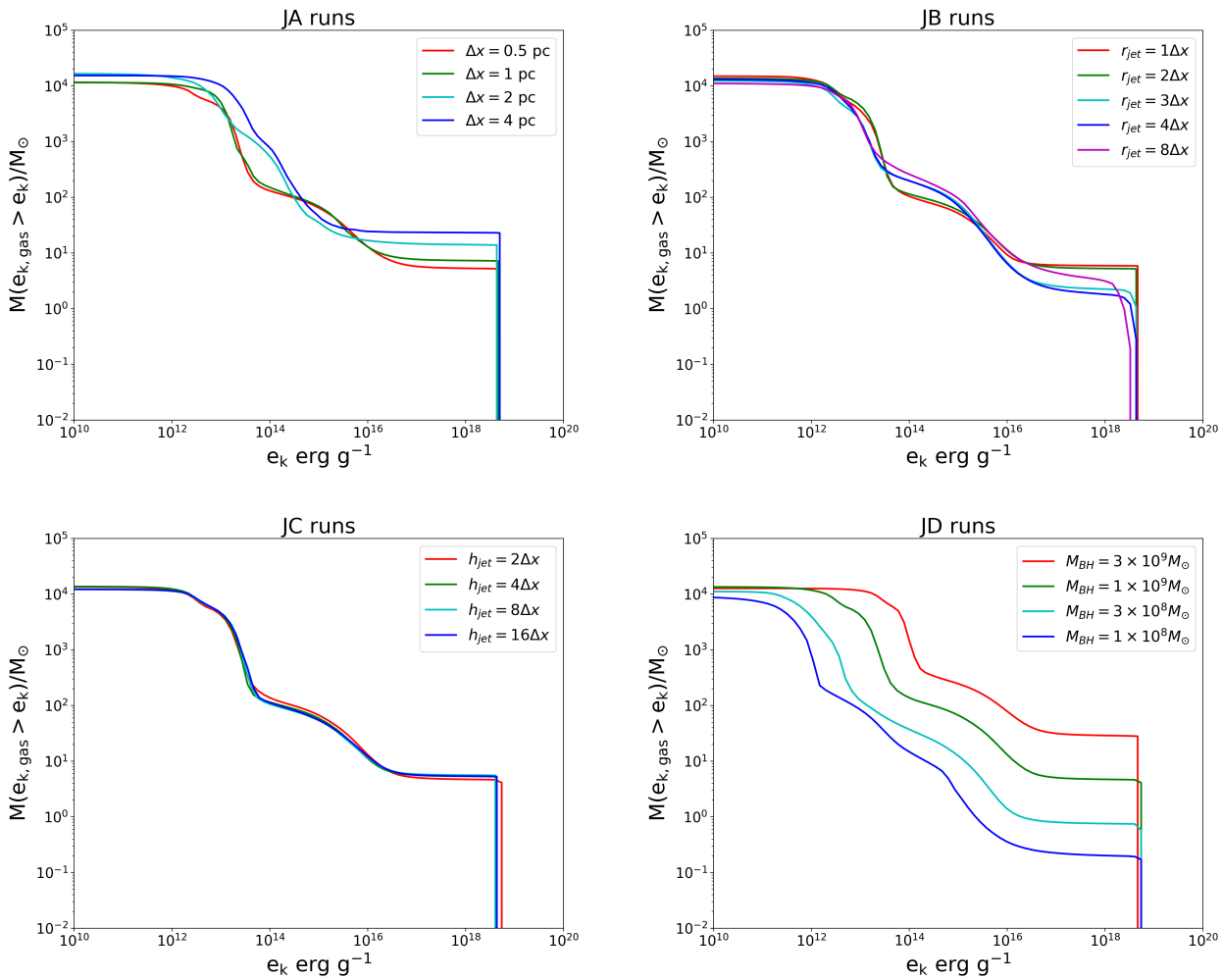

Figure 13. Cumulative kinetic energy distribution of entrained material for different runs (JA, JB, JC and JD).

to the Bondi radius to be different. In run JD4, the Bondi radius is around $7\Delta x$, which is smaller than the control surface radius. This provides an insight into how jet power and morphology change if the Bondi radius is less resolved when calculating the accretion rate.

Slice plots for the JD runs are shown in Figure 18. With decreasing central SMBH mass (thus decreasing Bondi radius), the jet becomes less powerful and the shock region smaller. The lower mass SMBHs also sustain a lower energy budget, despite reaching a steady state. We see in Figures 13 and 17 that a lower-mass central SMBH generates less entrained material, and the gas also has lower velocity. This indicates the jet power is relatively smaller for lower mass SMBHs. We notice, however, comparing to the inflow gas in each individual run, the jet power is not necessarily small. In Figure 15 we observe that the accretion rate measured by the incoming flow is in fact higher for the lower mass SMBH compared to the Bondi-Hoyle-Lyttleton rate, indicating that when measuring accretion outside of the Bondi zone, we tend to over-estimate the accretion rate and thus the jet power.

3.2.4. *Comparing with Bondi-Formula Accretion*

To compare our sink particle model with the traditional approach based on the Bondi formula, we ran a comparison run where the initial setup, jet model, boundary conditions, and reset criteria remain the same (using the most common configuration from run JC1), while only the method of accretion rate measurement is different. To determine the accretion rate in the "Bondi formula" run, we compute the sound speed and density by averaging zones lying within the control surface but not the jet injection cylinder. We still apply kernel resetting. To compare with similar simulations in the literature, we use the α-Bondi configuration, where $\alpha = 100$, to calculate the final accretion rate and thus